

Basic Mech. Engg.



Relative Motion Analysis: Velocity

POSITION, DISPLACEMENT, & VELOCITY

- For Position of B ,
$$\mathbf{r}_B = \mathbf{r}_A + \mathbf{r}_{B/A}$$
- During an instant of time dt , points A and B undergo displacements $d\mathbf{r}_A$ and $d\mathbf{r}_B$

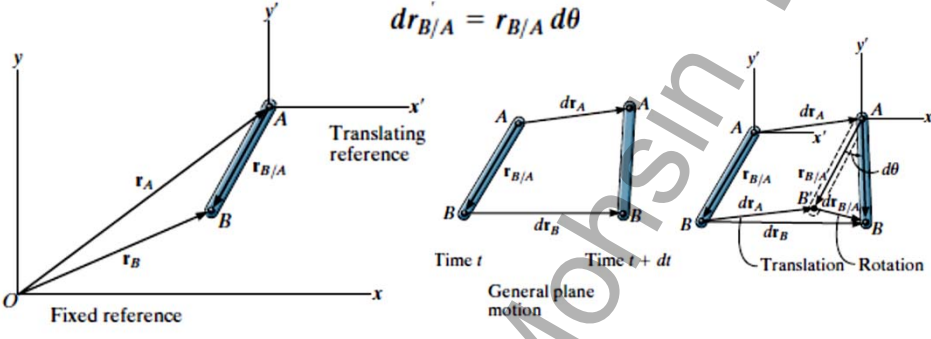
$$d\mathbf{r}_B = d\mathbf{r}_A + d\mathbf{r}_{B/A}$$

due to translation and rotation

due to translation of A

due to rotation about A

$$d\mathbf{r}_{B/A} = \mathbf{r}_{B/A} d\theta$$



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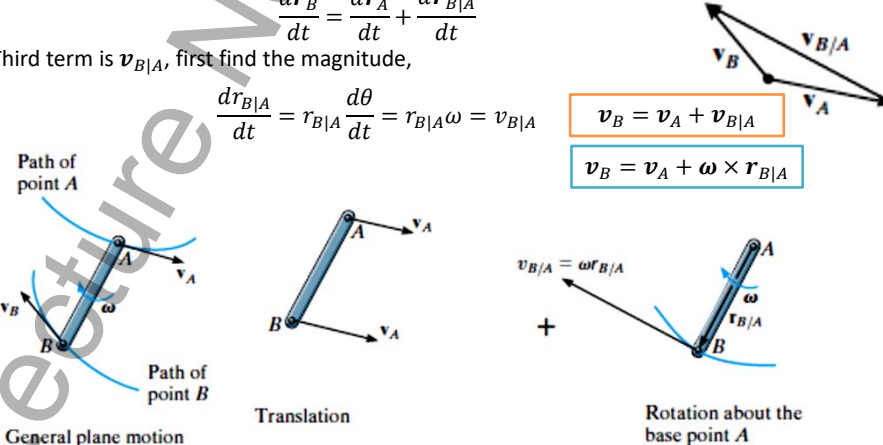
Relative Motion Analysis: Velocity

POSITION, DISPLACEMENT, & VELOCITY

- Relationship between velocities of points A and B , taking the first derivative
$$\frac{d\mathbf{r}_B}{dt} = \frac{d\mathbf{r}_A}{dt} + \frac{d\mathbf{r}_{B/A}}{dt}$$

Third term is $\mathbf{v}_{B/A}$, first find the magnitude,
$$\frac{dr_{B/A}}{dt} = r_{B/A} \frac{d\theta}{dt} = r_{B/A} \omega = v_{B/A}$$

$$\mathbf{v}_B = \mathbf{v}_A + \mathbf{v}_{B/A}$$
$$\mathbf{v}_B = \mathbf{v}_A + \boldsymbol{\omega} \times \mathbf{r}_{B/A}$$



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Procedure for Analysis: Relative Velocity



VECTOR ANALYSIS

- **Kinematic Diagram:**

- Establish the directions of the fixed x, y coordinates and draw a kinematic diagram of the body. Indicate on it the velocities of points A and B , ω and $r_{B|A}$.

- If the magnitudes of v_A , v_B , or ω are unknown, the sense of or direction of these vectors can be assumed.

- **Velocity Equation:**

- Represent all the vectors in Cartesian vector notation,

$$v_B = v_A + \omega \times r_{B|A}$$

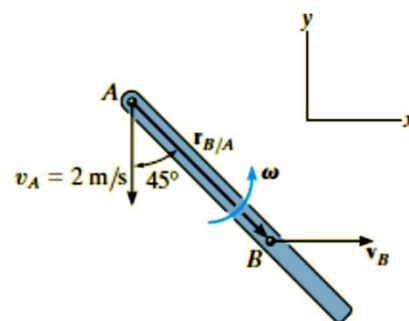
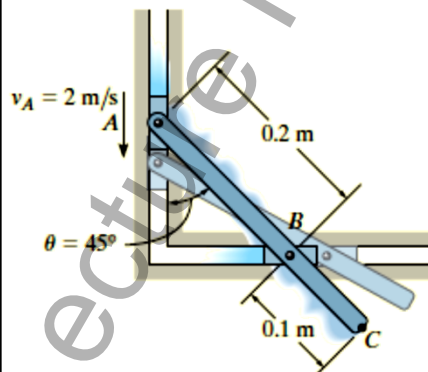
- If the answer is *negative*, it indicates the sense of direction of the vector is opposite to that shown in kinematic diagram.

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The link shown in Fig. 16–13a is guided by two blocks at A and B , which move in the fixed slots. If the velocity of A is 2 m/s downward, determine the velocity of B at the instant $\theta = 45^\circ$.



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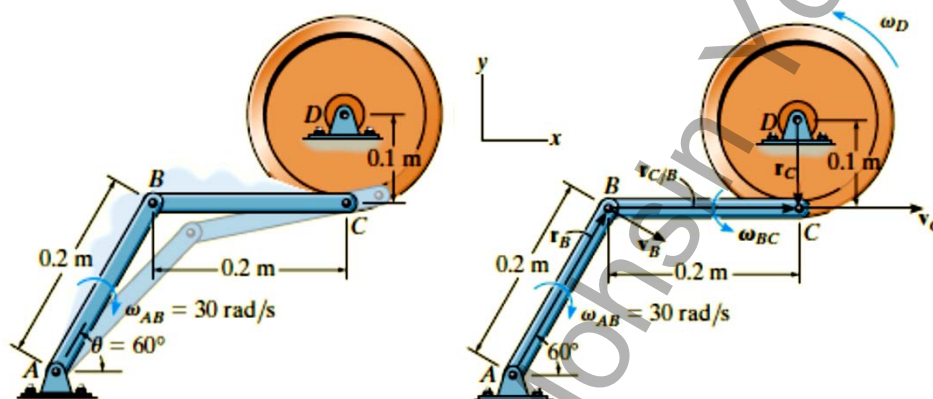
The cylinder shown in Fig. 16–14a rolls without slipping on the surface of a conveyor belt which is moving at 2 m/s. Determine the velocity of point A. The cylinder has a clockwise angular velocity $\omega = 15 \text{ rad/s}$ at the instant shown.

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The collar C in Fig. 16–15a is moving downward with a velocity of 2 m/s. Determine the angular velocity of CB at this instant.

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The bar AB of the linkage shown in Fig. 16-16a has a clockwise angular velocity of 30 rad/s when $\theta = 60^\circ$. Determine the angular velocities of member BC and the wheel at this instant.

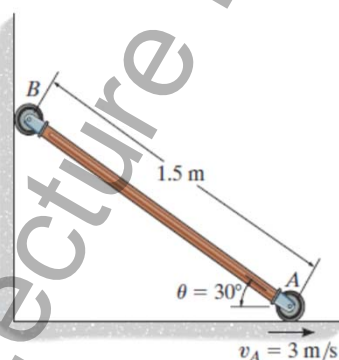


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F16-7. If roller A moves to the right with a constant velocity of $v_A = 3 \text{ m/s}$, determine the angular velocity of the link and the velocity of roller B at the instant $\theta = 30^\circ$.



$$\begin{aligned}
 \mathbf{v}_B &= \mathbf{v}_A + \boldsymbol{\omega} \times \mathbf{r}_{B/A} \\
 -v_B \mathbf{j} &= (3 \mathbf{i}) \text{ m/s} \\
 &\quad + (\omega \mathbf{k}) \times (-1.5 \cos 30^\circ \mathbf{i} + 1.5 \sin 30^\circ \mathbf{j}) \\
 -v_B \mathbf{j} &= [3 - \omega_{AB}(1.5 \sin 30^\circ)] \mathbf{i} - \omega(1.5 \cos 30^\circ) \mathbf{j} \\
 0 &= 3 - \omega(1.5 \sin 30^\circ) \quad (1) \\
 -v_B &= 0 - \omega(1.5 \cos 30^\circ) \quad (2) \\
 \omega &= 4 \text{ rad/s} \quad v_B = 5.20 \text{ m/s} \quad \text{Ans.}
 \end{aligned}$$

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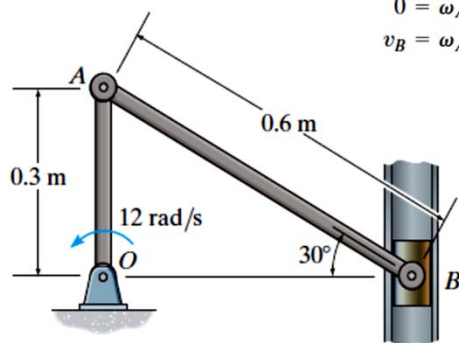
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F16-10. If crank OA rotates with an angular velocity of $\omega = 12 \text{ rad/s}$, determine the velocity of piston B and the angular velocity of rod AB at the instant shown.

$$\begin{aligned}\mathbf{v}_A &= \boldsymbol{\omega}_{OA} \times \mathbf{r}_A \\ &= (12 \text{ rad/s})\mathbf{k} \times (0.3 \text{ m})\mathbf{j} \\ &= [-3.6\mathbf{i}] \text{ m/s}\end{aligned}$$

$$\begin{aligned}\mathbf{v}_B &= \mathbf{v}_A + \boldsymbol{\omega}_{AB} \times \mathbf{r}_{B/A} \\ v_B\mathbf{j} &= (-3.6 \text{ m/s})\mathbf{i} \\ &\quad + (\omega_{AB}\mathbf{k}) \times (0.6 \cos 30^\circ\mathbf{i} - 0.6 \sin 30^\circ\mathbf{j}) \text{ m} \\ v_B\mathbf{j} &= [\omega_{AB}(0.6 \sin 30^\circ) - 3.6]\mathbf{i} + \omega_{AB}(0.6 \cos 30^\circ)\mathbf{j} \\ 0 &= \omega_{AB}(0.6 \sin 30^\circ) - 3.6 \quad (1) \\ v_B &= \omega_{AB}(0.6 \cos 30^\circ) \quad (2)\end{aligned}$$



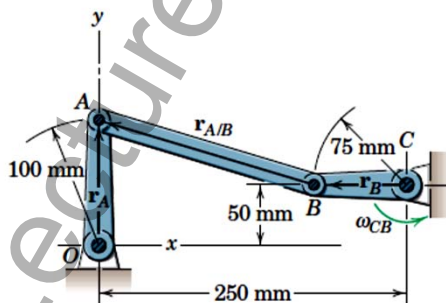
$$\omega_{AB} = 12 \text{ rad/s} \quad v_B = 6.24 \text{ m/s} \uparrow$$

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Crank CB oscillates about C through a limited arc, causing crank OA to oscillate about O . When the linkage passes the position shown with CB horizontal and OA vertical, the angular velocity of CB is 2 rad/s counterclockwise. For this instant, determine the angular velocities of OA and AB .



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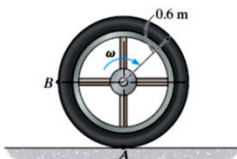
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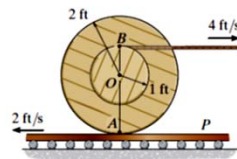
Relative Motion Analysis: Velocity

FUNDAMENTAL PRACTICE PROBLEMS

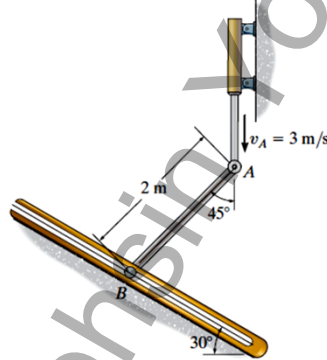
F16-8. The wheel rolls without slipping with an angular velocity of $\omega = 10 \text{ rad/s}$. Determine the magnitude of the velocity of point B at the instant shown.



F16-9. Determine the angular velocity of the spool. The cable wraps around the inner core, and the spool does not slip on the platform P .



F16-12. End A of the link has a velocity of $v_A = 3 \text{ m/s}$. Determine the velocity of the peg at B at this instant. The peg is constrained to move along the slot.



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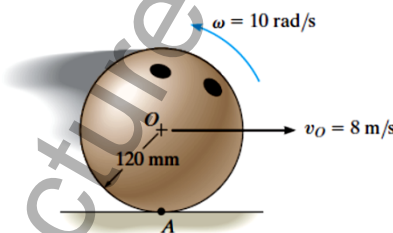
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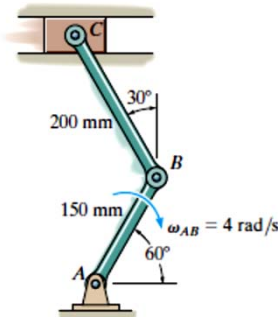
Relative Motion Analysis: Velocity

PRACTICE PROBLEMS

16-58. A bowling ball is cast on the "alley" with a backspin of $\omega = 10 \text{ rad/s}$ while its center O has a forward velocity of $v_O = 8 \text{ m/s}$. Determine the velocity of the contact point A in contact with the alley.



***16-68.** If bar AB has an angular velocity $\omega_{AB} = 4 \text{ rad/s}$, determine the velocity of the slider block C at the instant shown.



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